# KRISHNA ANVITH VATTIKUTI

Backend Software Engineer | Python, APIs, LLM-Integrated Systems

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## SUMMARY

Entry-level Backend Software Engineer with hands-on experience designing and deploying Python-based services, including FastAPI backends, SQLite/SQLAlchemy data layers, and containerized applications with Docker. Experience building and integrating LLM-powered features into production-style systems, including Retrieval-Augmented Generation (RAG) pipelines, agent orchestration, and observability tooling. Strong foundation in system design, API development, and classical machine learning from applied internships, with 97.5% model accuracy delivered in production-style pipelines.

## TECHNICAL SKILLS

**Backend & Deployment:** Python, FastAPI, REST APIs, Docker, SQLite/SQLAlchemy, Git, System Design Basics, CI/CD Basics

**Languages & Data:** Python, SQL, Pandas, NumPy, Matplotlib, Tableau, Power BI

**LLM & Applied GenAI:** RAG, LangChain, ChromaDB, Hugging Face Transformers, Ollama, Llama-3, QLoRA/PEFT Fine-Tuning, Prompt Engineering

**Agents & Automation:** ReAct Agents, Model Context Protocol (MCP), Tool Calling, Playwright

**Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Feature Engineering, Model Evaluation, TensorFlow Lite

## PROJECTS

**AgentWatch - Observability & Evaluation Platform for AI Agents** (Github-Repository) Jun 2026

- Designed and built a FastAPI backend, SQLite/SQLAlchemy storage layer, and a Python SDK to capture agent runs, spans, latency, inputs, outputs, statuses, and evaluation results at scale.
- Built a React + Vite dashboard on top of the backend API to let engineers monitor runs, inspect tool calls, and debug agent failures end-to-end.
- Implemented rule-based evaluation logic in the backend service to automatically flag failed runs, slow responses, empty outputs, and tool-execution errors across simulated agent workflows.

**RAFT ArXiv - Retrieval-Augmented Fine-Tuning** (Github-Repository) Mar 2026

- Built a FastAPI service exposing a side-by-side evaluation UI to benchmark RAFT vs. Llama-3 outputs on citation precision, answer faithfulness, and refusal accuracy.
- Fine-tuned Phi-2 (2.7B) with QLoRA on a custom ArXiv QA dataset built from 200 papers, then wrapped the model behind an API for citation-grounded answer generation.
- Created a custom RAFT training dataset with an 80/20 oracle-to-distractor context ratio, teaching the model to cite relevant passages and refuse unsupported answers.

**MCP Agent Browser - LLM Browser Automation Agent** (Github-Repository) Nov 2025

- Built a Python service using LangChain ReAct planning and Playwright to convert natural-language instructions into executable actions (click, fill, navigate, extract) through a structured tool-calling interface.
- Validated the system across 20+ real-world workflows and ran failure analysis on ambiguous instructions, prompt structure, tool execution, and multi-step planning.
- Improved system reliability through iterative refinement of the tool-calling and orchestration layer, and documented workflow-level failure patterns.

## TECHNICAL EXPERIENCE

**UpGrad - Data Science Intern** Jan 2024 – May 2024

- Developed a TensorFlow training pipeline for a Human Activity Recognition system across 6 activity classes, modularizing preprocessing and feature extraction into 4 stages to support faster model iteration.
- Applied ensemble classification and threshold tuning on accelerometer/gyroscope time-series data, reaching 97.5% accuracy and improving fall-detection reliability.
- Improved feature engineering and model iteration workflows, reducing preprocessing time by 30% and model-training time by 20%.

**TekWissen Software (Computer Vision & IoT Summer Intern)** May 2023 – Aug 2023

- Built an end-to-end computer vision pipeline for real-time plant disease detection using TensorFlow and OpenCV across 8 disease classes, optimizing preprocessing and inference for sub-second latency on edge hardware.
- Converted the trained model to TensorFlow Lite with post-training quantization for IoT deployment on resource-constrained devices.
- Built an automated data pipeline generating 5,000+ synthetic training samples weekly to address underrepresented classes.

## EDUCATION

**New Jersey Institute of Technology - M.S. Data Science, Newark, NJ** Aug 2024 – May 2026

**Lovely Professional University - B.Tech Computer Science, Punjab, India** Jul 2020 – Jul 2024